



Anjali Wadhe

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B.Tech

Metallurgical Engineering and Material Sciences

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EDUCATION

- Awarded with **FFE Scholar** Fellowship given for advancing studies in Engineering and Technology [2023]
- Secured **Dakshana Scholarship**, provided to **2%** students among **25k+** students across 625 JNVs [2022]
- Achieved **1st rank** in the seminar, during **Regional Integration Meet, NVS** among **60+** students [2018]

KEY PROJECT

Alloy Design Tool | Supervised Learning Project [July'24 - Present]

- Developed a **Django** web tool integrating **pycalphad** for thermodynamic alloy property and melt pool calculations also designed a friendly user interface on Figma and successfully integrated it with frontend
- Used **Nelder-Mead** optimization to refine Fe-Mn-C alloy parameters and enhance property predictions
- Installed **Scheil Model** to get Retained Austenite Fraction and Phase fraction during martensite formation

COURSE PROJECTS

UI for Time management App | Design Thinking and Innovation [Jan'24-April'24]

- Analyzed student's daily schedules and designed an app prototype on **Figma** to address the key challenges
- Presented the app concept to an external panel, showcasing data-driven insights and innovative solutions

Pill Dispenser | Tech-Opportunities and Identification [Jan'24-Apr'24]

- Pitched a pill dispenser for medication management with automated dose reminders and refill assistance
- Collaborated in a team of 3 to conduct a market research on existing solutions and their USP, competitors, market size, risk and challenges and revenue models to develop a data-driven business strategy

SELF PROJECTS

Object Detection | WIDS | Analytics Club, IIT Bombay [Dec'23-Jan'24]

- Developed and implemented an object detection system using **YOLOv3**, achieving high accuracy and real-time processing capabilities for the identification and localization of multiple objects within images
- Applied advanced data preprocessing techniques contributing to an impactful solution for **CV** applications

Demon Slayer AI: Mastering Doom with RL | SOC'24 | IIT Bombay [May'24-July'24]

- Developed a deep understanding of **RL** algorithms and effectively applied them to build an AI agent
- Engineered an **AI agent** that dynamically **interacts** with its environment to optimize its performance
- Designed agent to strategically engage and overcome adversaries, showcasing practical integration of **RL**

AI/ML in Material Science | SOS'24 | IIT Bombay [May'24-July]

- Deployed ML models for material properties predictions, applying AI/ML techniques to material science
- Earned IBM's Deep Learning Specialization, gaining expertise in neural networks, deep learning techniques
- Applied deep learning methods to diverse domains, including material science, to enhance analysis

POSITION OF RESPONSIBILITY

Overall Coordinator | Kaladarshan'25 | Rang and Pixels, IIT Bombay [Jan'25-Present]

- Oversaw the successful implementation of 4 key event segments and 2 day exhibition, ensuring smooth operations and high engagement for one of the institute's premier art and photography exhibitions
- Managed cross-functional coordination, logistics, and enhancing participation and outreach for the event

TECHNICAL SKILLS AND INTERESTS

Languages: Python (Libraries: NumPy, Pandas, PyTorch, SciPy, Matplotlib, Tensorflow), MATLAB, C++

Developer Tools/Frameworks: HTML/CSS, Django, Git/GitHub, AutoCAD, MS Office

Softwares : ThermoCalc, Photoshop, Illustrator, Figma, LTSPice, Arduino IDE

EXTRA CURRICULAR SKILLS & CERTIFICATIONS

- Earned certification in **HTML, CSS, and JavaScript** from YouAccel Training, gained expertise in web-dev
- Attended a two-year long course in **Fine Arts** conducted by National Sports Organization(NSO)
- Participated in **Cultconnect** Showcasing Favourite Place in Institution conducted by **Design Club**
- Created a **News Website** as a Recruitment project for **Devcom**(Developer's Community IIT Bombay)
- Completed 5 levels of **Abacus** and 3 levels of **Vedic Mathematics**, enhancing numerical proficiency